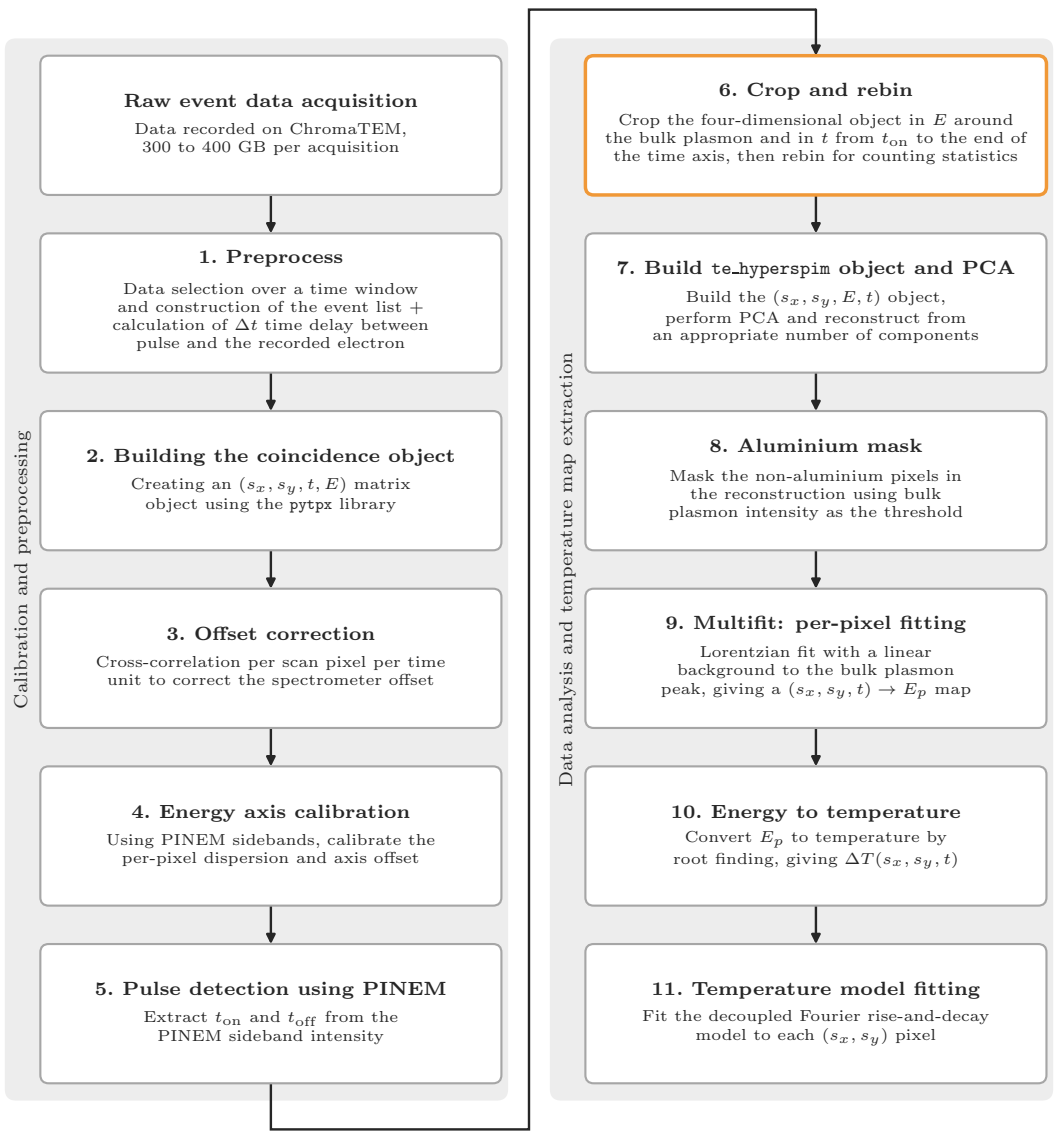

Raw event data acquisition
Data recorded on ChromaTEM, 300 to 400 GB per acquisition
1. Preprocess
Data selection over a time window and construction of the event list + calculation of Δt time delay between pulse and the recorded electron
2. Building the coincidence object
Creating an (s_x, s_y, t, E) matrix object using the pytpx library
3. Offset correction
Cross-correlation per scan pixel per time unit to correct the spectrometer offset
4. Energy axis calibration
Using PINEM sidebands, calibrate the per-pixel dispersion and axis offset
5. Pulse detection using PINEM
Extract t_{on} and t_{off} from the PINEM sideband intensity
Calibration and preprocessing
6. Crop and rebin
Crop the four-dimensional object in E around the bulk plasmon and in t from t_{on} to the end of the time axis, then rebin for counting statistics
7. Build te_hyperspim object and PCA
Build the (s_x, s_y, E, t) object, perform PCA and reconstruct from an appropriate number of components
8. Aluminium mask
Mask the non-aluminium pixels in the reconstruction using bulk plasmon intensity as the threshold
9. Multifit: per-pixel fitting
Lorentzian fit with a linear background to the bulk plasmon peak, giving a $(s_x, s_y, t) \to E_p$ map
10. Energy to temperature
Convert E_p to temperature by root finding, giving $\Delta T(s_x, s_y, t)$
11. Temperature model fitting
Fit the decoupled Fourier rise-and-decay model to each (s_x, s_y) pixel
Data analysis and temperature map extraction